

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 11 END OF COURSE EXAMINATION

MATHEMATICS

2005

Time allowed: 120 minutes

Directions to Candidates

- Attempt all questions
- Start each question on a new page
- All necessary working should be shown
- Unless otherwise specified, answers must be given in their simplest form
- Approved calculators may be used in all sections.
- Use a ruler when drawing straight lines
- Marks may be deducted for careless or poorly arranged work.
- Marks shown are approximate and may be varied.
- These questions must be attached to the top of your solutions.

Name: _____ Class: _____

1	2	3	4	5	6	7	8	TOTAL
/12	/13	/13	/13	/12	/12	/13	/12	/100

QUESTION 1: (12 Marks)

- 1 (a) Express $2\sqrt{3} \times \sqrt{27}$ as an integer
- 1 (b) (i) If $v^2 = u^2 + 2as$, find the value of a if $v = 5$, $u = -2$ and $s = 10$.
- 1 (ii) Write 23.8249 correct to 2 decimal places
- 2 (c) (i) Fully factorise $16x^2 - 4$
- 1 (ii) Fully factorize $6m^2 - 5m - 4$
- 2 (d) Simplify $\frac{2x-y}{4} - \frac{x+y}{6}$
- 2 (e) Find $\lim_{x \rightarrow 3} \left(\frac{x^2 - 9}{x - 3} \right)$
- 2 (f) Express $\frac{2}{\sqrt{5} - \sqrt{3}}$ with a rational denominator

QUESTION 2: (13 Marks)

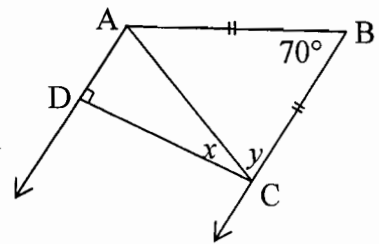
- 3 (a) Solve $x(x-2) = 2$ leaving your answer as a surd in simplest form.
- 3 (b) Solve the following inequality and plot your answer on a number line:

$$|x-2| \geq 4$$

- 2 (c) Solve $x^2 < 3x$
- 3 (d) In the diagram at right, you are given that

$$\begin{aligned} AD &\parallel BC \\ AB &= BC \\ \angle ABC &= 70^\circ \\ AD &\perp DC \end{aligned}$$

SHOWING ALL WORKING, and giving all reasons, calculate the size of $\angle ACD$



- 2 (e) Prove that $\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta = \tan \theta$

QUESTION 3: (13 Marks)

- 1 (a) (i) Find the midpoint, **R**, of the line joining the points A(3, 5) to B(7, -3)
- 3 (ii) Find the equation of the line through **R** in part (i) and parallel to the line $3y + 2x = 7$. Give your answer in general form.
- (b) For the parabola $y = 2x^2$
- 3 (i) Find the equation of the tangent to the curve at the point P where $x = -1$
- 3 (ii) Find the equation of the normal to the curve at the point P where $x = -1$
- 3 (iii) The tangent and the normal above cut the y-axis at T and N respectively. Find the length of TN.

QUESTION 4: (13 Marks)

- (a) Differentiate with respect to x :
- 3 (i) $y = 3x^2 + 2x - \frac{5}{x}$
- 1 (ii) $y = 3\sqrt{x}$
- 1 (iii) $y = (3x^2 - 1)^5$
- 2 (iv) $y = \frac{5}{x-2}$
- 2 (v) $y = \frac{x-1}{2x+3}$ (Give answer in simplest form)
- 1 (b) If $f(x) = 5x^2 + 2$, find $f'(2)$
- 3 (c) If $y = (x^2 - x + 1)^3$, show that $x = \frac{1}{2}$ is the ONLY solution to $\frac{dy}{dx} = 0$

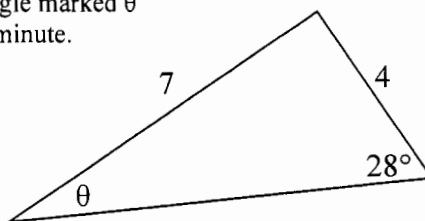
QUESTION 5: (12 Marks)

- 3 (a) State the largest possible domain for the function $f(x) = \sqrt{25 - x^2}$
- 2 (b) Determine if the function $f(x) = \frac{1}{x^2 - 4}$ is odd, even or neither, showing clearly all reasoning
- 3 (c) What is the equation of the circle which touches the x -axis at $(2, 0)$ and the y -axis at $(0, 2)$
- 4 (d) Sketch the function $y = f(x)$ which is defined by :

$$f(x) = \begin{cases} x - 2, & x \leq 2 \\ (x - 2)^2, & x > 2 \end{cases}$$

QUESTION 6: (12 Marks)

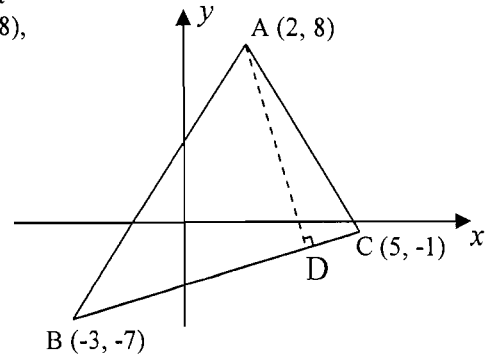
- 2 (a) Find the equation of the line which passes through the point $(3, -1)$ and makes equal intercepts on the two positive co-ordinate axes.
- 3 (b) Find the size of the angle marked θ correct to the nearest minute.



- (c) Two adjacent sides of a parallelogram have sides of 8 cm and 10cm, while the longer diagonal is 14 cm
- (i) Draw the diagram neatly onto your answer sheet (NO MARKS)
- 2 (ii) Calculate the size of the larger angle of the parallelogram, correct to the nearest minute.
- 3 (iii) Calculate the length of the shorter diagonal correct to 2 decimal places.
- 2 (d) Give the exact value of $2\sin 45^\circ \cos 60^\circ$ giving your answer with a rational denominator.

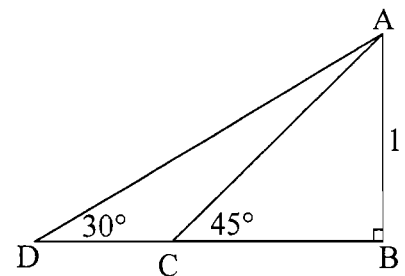
QUESTION 7: (13 Marks)

- (a) The triangle ABC shown at right, has vertices of A(2, 8), B(-3, -7) and C(5, -1). AD is an altitude.



- 1 (i) Find the length of BC
2 (ii) Find the equation of BC
2 (iii) Find the distance of A to BC
1 (iv) Find the area of $\triangle ABC$

- 3 (b) (i) In the diagram at right, find the exact length of DC
1 (ii) Find the size of $\angle DAC$
3 (iii) Hence prove, using the figure given, that the exact value of $\sin 15^\circ$ is $\frac{\sqrt{3}-1}{2\sqrt{2}}$



QUESTION 8: (12 Marks)

- 3 (a) Find $\frac{d}{dx} [x^2(1-2x)^3]$ (Give answer in simplest factored form)
- 5 (b) Find the equation of the normal to the semi-circle $y = \sqrt{100 - x^2}$ at the point (6, 8) on it. Give the equation in general form.
- 4 (c) Solve the equation $4^x - 3 \cdot 2^x - 4 = 0$

Q 1

$$\sqrt{27} = 2\sqrt{3} \times 3\sqrt{3}$$

$$= 18$$

$$5 = 4 + 20a$$

$$1 = 20a$$

$$= \frac{21}{20}$$

$$23 \cdot 82$$

$$= 18$$

$$(4x^2 - 1) = \frac{4(2x-1)(2x+1)}{2m \times 4 + 1}$$

$$\frac{x-y}{4} = \frac{x+y}{6}$$

$$3(2x-y) - 2(2x+y)$$

$$6x - 3y - 2x - 2y$$

$$4x - 5y$$

$$= \frac{6}{12}$$

$$= \frac{(x/3)(x+3)}{(x-3)}$$

$$= \frac{2}{-3} \times \frac{\sqrt{5} + \sqrt{3}}{\sqrt{5} + \sqrt{3}}$$

$$= \frac{2\sqrt{5} + 2\sqrt{3}}{2}$$

$$= \sqrt{5} + \sqrt{3}$$

$$= 2$$

$$= 2$$

$$= 2$$

$$= 2$$

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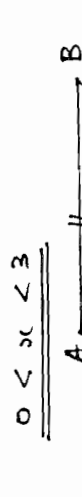
b) $|x-2| \geq 4$

$$x-2 \geq 4 \quad x-2 \leq -4$$

$$x \geq 6 \quad x \leq -2$$



c) $x^2 - 3x < 0$
 $x(x-3) < 0$
 $0 < x < 3$



$y = 55^\circ$ (opposite equal sides in isosceles triangle)

$x = 35^\circ$ (interior angles $AB \parallel BC$)

e) LHS = $\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cdot \cos \theta$

$$= \frac{\sin^3 \theta + \sin \theta \cos^2 \theta}{\cos \theta}$$

$$= \frac{\sin \theta (\sin^2 \theta + \cos^2 \theta)}{\cos \theta}$$

$$= \frac{\sin \theta}{\cos \theta}$$

$$= \tan \theta$$

$$= \text{RHS}$$

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QUESTION 3

a) i) $R(5,1)$

ii) $3y + 2x = 7$

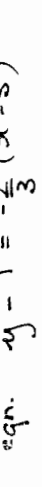
$$y = -\frac{2x}{3} + \frac{7}{3}$$

$$\therefore m = -\frac{2}{3}$$

eqn. $xy - 1 = -\frac{2}{3}(x-5)$

$$3y - 3 = -2x + 10$$

$$2x + 3y - 13 = 0$$



b) i) $\frac{dy}{dx} = 4x$

eqn of tang: $y-2 = -4(x+1)$

$$y = -4x - 2$$

ii) eqn of normal: $m = \frac{1}{4}$

$$y-2 = \frac{1}{4}(x+1)$$

$$4y-8 = x+1$$

$$x-4y+9=0$$

iii) $T(0, -2) \therefore TN = 4\frac{1}{4}$ units

$N(0, \frac{9}{4})$

QUESTION 4

a) i) $\frac{dy}{dx} = 6x + 2 + 5x^{-2}$

$$= 6x + 2 + 5x^{-2}$$

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$$= 6x + 2 + 5x^{-2}$$

$$= 6x + 2 + 5x^{-2}$$

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ii) $y = 3x^{1/2}$

$$\frac{dy}{dx} = \frac{3}{2}x^{-1/2}$$

$$= \frac{3}{2\sqrt{x}}$$

iii) $\frac{dy}{dx} = 5 \cdot 6x(3x^2-1)^4$

$$= 30x(3x^2-1)^4$$

iv) $y = 5(x-2)^{-1}$

$$\frac{dy}{dx} = -5(x-2)^{-2} = \frac{-5}{(x-2)^2}$$

v) $y = \frac{x-1}{2x+3}$

$$u = x-1 \quad v = 2x+3$$

$$u' = 1 \quad v' = 2$$

$$\frac{dy}{dx} = \frac{2x+3 - 2(x-1)}{(2x+3)^2}$$

$$= \frac{5}{(2x+3)^2}$$

b) $f(x) = 5x^2 + 2$

$$f'(x) = 10x$$

$$f'(2) = 20$$

c) $y = (x^2-x+1)^3$

$$\frac{dy}{dx} = 3(2x-1)(x^2-x+1)^2$$

solve $3(2x-1)(x^2-x+1) = 0$

$$2x-1=0 \quad x^2-x+1=0$$

$$x = \frac{1}{2} \quad \Delta = 1-4 \times 1 \times 1$$

$$\Delta = -3$$

$$\therefore \text{no solution}$$

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ION 5

$$-x^2 \geq 0$$

$$1)(5+x) \geq 0$$

$$5x \leq 5$$

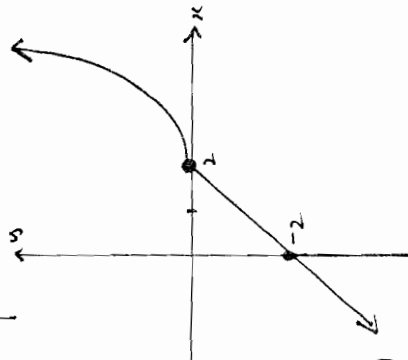
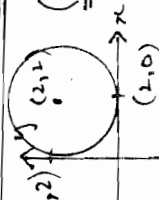
$$f(x) = \frac{1}{x^2 - 4}$$

$$f(-x) = \frac{1}{(-x)^2 - 4}$$

$$= \frac{1}{x^2 - 4}$$

$$f(x) = f(-x) \text{ EVEN}$$

$$(x-2)^2 + (y-2)^2 = 4$$



ION 6

Let line cut
x & y axes at
a

$$\therefore m = -1$$

$$(3, -1)$$

$$\text{eqn: } y+1 = -(x-3)$$

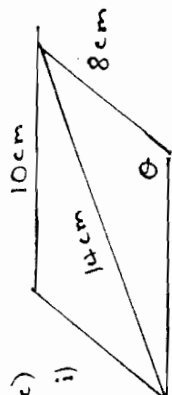
$$y+1 = -x+3$$

$$x+y-2=0$$

$$\sin \theta = \frac{\sin 28^\circ}{4}$$

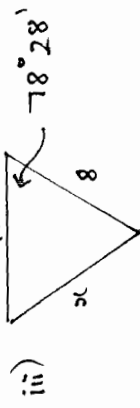
$$\sin \theta = \frac{4 \sin 28^\circ}{7}$$

$$\theta = 15^\circ 34'$$



$$\cos \theta = \frac{8^2 + 10^2 - 14^2}{2 \times 8 \times 10}$$

$$\theta = 101^\circ 32'$$



$$x^2 = 10^2 + 8^2 - 2 \times 10 \times 8 \cos 78^\circ 28'$$

$$x = 11.49 \text{ cm}$$

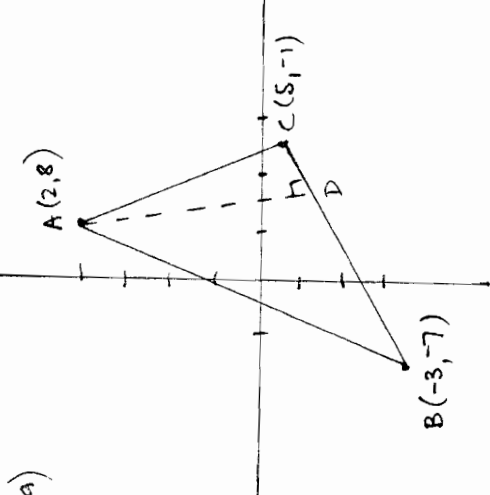
$$d) 2 \sin 45^\circ \cos 60^\circ$$

$$2 \times \frac{1}{\sqrt{2}} \times \frac{1}{2}$$

$$\frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{2}$$

$$\frac{\sqrt{2}}{2}$$

QUESTION 7



$$i) BC = \sqrt{(5+3)^2 + (-1+7)^2}$$

$$= \sqrt{64 + 36}$$

$$= 10 \text{ units}$$

$$ii) m = \frac{-1+7}{5+3} = \frac{6}{8} = \frac{3}{4}$$

$$\text{eqn } y+1 = \frac{3}{4}(x-5)$$

$$4y+4 = 3x-15$$

$$0 = 3x-4y-19$$

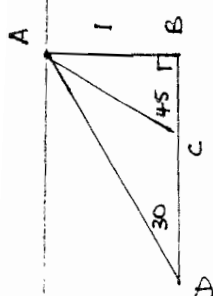
$$iii) p = \left| \frac{3 \times 2 - 4 \times 8 - 19}{\sqrt{9+16}} \right|$$

$$= \left| \frac{-45}{5} \right|$$

$$= 9 \text{ units}$$

$$v) \text{ Area } \triangle ABC = \frac{1}{2} \times 10 \times 9$$

b) i)



$BC = 1$ (isosceles $\triangle$)

$$\tan 30^\circ = \frac{1}{DB}$$

$$\frac{1}{\sqrt{3}} = \frac{1}{DB}$$

$$\therefore DB = \sqrt{3}$$

$$\therefore DC = (\sqrt{3}-1) \text{ units}$$

$$ii) \hat{AC} = 15^\circ \text{ (ext angle of } \triangle)$$

$$iii) AC = \sqrt{2} \text{ units}$$

$$\text{Rule: } \frac{\sin 15^\circ}{\sqrt{3}-1} = \frac{\sin 30^\circ}{\sqrt{2}}$$

$$\sin 15^\circ = \frac{(\sqrt{3}-1) \cdot \frac{1}{2}}{\sqrt{2}}$$

$$= \frac{(\sqrt{3}-1)}{2\sqrt{2}}$$

QUESTION 8

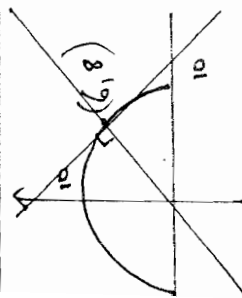
$$a) u = x^2 \quad v = (1-2x)^3$$

$$u' = 2x \quad v' = -6(1-2x)^2$$

$$\frac{dy}{dx} = 2x(1-2x)^3 - 6x^2(1-2x)^2$$

$$= 2x(1-2x)^2[(1-2x) - 3x]$$

$$= 2x(1-2x)^2(1-5x)$$



$$\sqrt{100 - x^2}$$

$$(100 - x^2)^{\frac{1}{2}} = \frac{1}{2} x - 2x(100 - x^2)^{-\frac{1}{2}}$$

$$= \frac{-x}{\sqrt{100 - x^2}}$$

$$= \frac{-6}{\sqrt{100 - 36}} = \frac{-6}{8} = -\frac{3}{4}$$

$$\text{normal: } y - 8 = \frac{4}{3}(x - 6)$$

$$3y - 24 = 4x - 24$$

$$\underline{4x - 3y = 0}$$

$$x^2 - 3 \cdot 2^x - 4 = 0$$

$$u = 2^x$$

$$u^2 - 3u - 4 = 0$$

$$(u - 4)(u + 1) = 0$$

$$u = 4 \quad u = -1$$

$$2^x = 4 \quad 2^x = -1$$

$$x = 2, \quad x = 0$$